Robotics and control systems

Precision Vehicle Tracking and Disturbance Recovery

Reference-Scheduled LPV-MPC

A theoretical formulation and implementation-grounded experimental study

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Research question

How does a constrained predictive controller recover from initial tracking errors and a temporary unmodeled disturbance while respecting steering, acceleration, and command-rate limits?

Abstract

This report develops the mathematical formulation of a vehicle-tracking study using a reference-scheduled linear parameter-varying model predictive controller (LPV-MPC). A nonlinear kinematic bicycle model is expressed in a moving reference frame, linearized about a feasible reference motion, and discretized to form a sequence of prediction models. A sparse quadratic program balances tracking error, deviation from feedforward commands, and command smoothness over a two-second horizon. The evaluation covers a variable-speed oval, a figure-eight, and a lane-change maneuver with an unmodeled lateral drift and longitudinal load. Saved experiments contain 1,520 successful optimization steps and full-run position RMSE between 0.289 and 0.396 m. The report derives the formulation, explains its distinction from the supplied six-state reference study, and establishes which conclusions follow from the numerical evidence. The contribution is a formulation and controlled evaluation of constrained tracking and recovery; no new solver or formal robustness theorem is claimed.

Builds on concepts from Mark Misin's study.

Contents

1 Research objective and project contribution 1

2 Notation, assumptions, and system architecture 2

3 Nonlinear vehicle and feasible reference motion 3

4 Derivation of moving-frame tracking errors 4

5 Reference-scheduled linear prediction model 5

6 Physical limits and horizon constraints 6

7 Objective function and tuning interpretation 7

8 Sparse quadratic-program construction 8

9 Optimization, feedback, and numerical acceptance 9

10 Local control properties and limits of the theory 10

11 Experiment design and performance measures 11

12 Trajectory tracking results 12

13 Error transients and interpretation 13

14 Disturbance response and recovery analysis 14

15 Command limits and optimization outcomes 15

16 Verification and reproducibility 16

17 Comparison, contribution, and research extensions 17

A Worked local model and QP dimensions 18

B Optimality conditions and theoretical interpretation 19

C Implemented components and correspondence to the formulation 20

References and evidence provenance 21

1 Research objective and project contribution

The broad goal is vehicle trajectory tracking: the vehicle should reach a prescribed position, heading, and speed at each instant. This study places particular emphasis on *recovery*: the vehicle begins away from its reference motion and, in one experiment, encounters disturbances absent from its prediction model. The controller must reduce the resulting errors while obeying limits on commands and their sample-to-sample changes.

The concrete research question is whether the chosen reference-scheduled prediction model and constrained horizon optimization can produce accurate, smooth recovery on the specified nonlinear plant. Its answer is evaluated through full-run errors, sampled constraint satisfaction, optimization success, and a separately defined post-disturbance recovery criterion.

1.1 Implemented contributions

1. **Moving-frame error formulation.** Along-track, cross-track, heading, and speed errors describe deviation from a time-indexed trajectory. This makes the tracking objective explicit in the coordinates relevant to the maneuver.
2. **Reference-scheduled prediction.** A local linear model changes with the known reference speed and curvature throughout the horizon. The controller anticipates the upcoming maneuver instead of freezing one model for the complete experiment.
3. **Sparse constrained optimization.** Predicted errors and absolute commands are optimization variables. Dynamics, amplitude bounds, command-rate bounds, and predicted-speed bounds are assembled into a quadratic program (QP).
4. **Separate nonlinear evaluation.** A nonlinear kinematic plant is integrated with fourth-order Runge–Kutta (RK4), while prediction uses a first-order discrete model. Initial offsets and an injected disturbance test feedback recovery.

1.2 Relationship to the earlier study

The supplied Mark Misin reference implements the same general receding-horizon idea with a six-state dynamic vehicle model. The present controller uses a four-state kinematic plant, a moving-frame error model, a sparse QP, and different experiment definitions. Those choices establish a distinct implementation and evaluation question. They do not by themselves establish a new MPC algorithm or better performance than the reference.

The theoretical background is conventional constrained MPC [1]; the numerical optimization is provided by OSQP [2]. The derivations and numerical details below follow the current local implementation and saved experiment records, rather than a generic controller template.

2 Notation, assumptions, and system architecture

Symbol	Meaning	Unit
$\boldsymbol{z} = [x, y, \psi, v]^\top$	Vehicle position, heading, and speed	m, rad, m/s
$\boldsymbol{u} = [\delta, a]^\top$	Front-wheel steering and acceleration command	rad, m/s^2
L	Wheelbase, fixed at 2.7	m
$(\cdot)_r$	Reference quantity at the same time as the state	varies
κ_r, ω_r	Reference curvature and yaw rate	m^{-1}, rad/s
$\boldsymbol{e} = [e_s, e_y, e_\psi, e_v]^\top$	Along-track, cross-track, heading, speed errors	varies
T_s, N	Sample period 0.1 s; prediction horizon 20	s, steps
$\boldsymbol{e}_{i\|k}, \boldsymbol{u}_{i\|k}$	Predictions i steps ahead, computed at sample k	varies
d_y, d_a	Global lateral drift and longitudinal load	m/s, m/s^2
Q, R, S, P_f	Error, feedforward, rate, terminal weights	scaled weights
$\boldsymbol{\xi}$	Stacked QP decision vector	mixed

Assumptions. The reference is analytic and available beyond the end of each run, so every optimization receives a complete preview. Reference speed is nonzero. The controller receives the exact simulated state; sensor noise, estimation errors, latency, and actuator lag are absent. Steering and acceleration commands are held constant during each control interval. The plant is planar and kinematic, with no tire-slip dynamics.

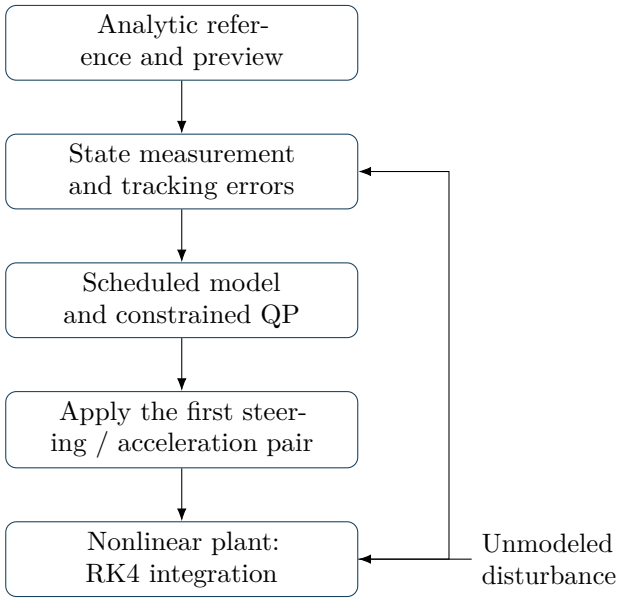


Figure 1: Receding-horizon feedback structure. Disturbances enter the plant and are not supplied to the controller.

Time-indexed tracking differs from geometric path following. A vehicle can lie on the correct curve yet have nonzero along-track error because it arrives too early or too late. The objective here penalizes that timing-related error.

3 Nonlinear vehicle and feasible reference motion

3.1 Kinematic bicycle plant

The nominal model maps vehicle speed and steering to planar translation and rotation:

$$\dot{\boldsymbol{z}} = \begin{bmatrix} v\cos\psi \\ v\sin\psi + d_y(t) \\ (v/L)\tan\delta \\ a + d_a(t) \end{bmatrix}. \tag{3.1}$$

The yaw-rate relation follows from a turning radius $R_c = L/\tan\delta$, giving $\dot{\psi} = v/R_c$. It assumes no lateral slip at the modeled reference point. The injected d_y is an additive drift in the global y direction; it is not a tire-force model. Likewise, d_a is an additive acceleration disturbance, not an identified powertrain load.

3.2 Reference geometry and feedforward commands

Let $\boldsymbol{p}_r(t) = [x_r(t), y_r(t)]^\top$ be twice differentiable with $v_r > 0$. The implementation calculates

$$v_r = \sqrt{\dot{x}_r^2 + \dot{y}_r^2}, \qquad \psi_r = \operatorname{atan2}(\dot{y}_r, \dot{x}_r), \tag{3.2}$$

$$\kappa_r = \frac{\dot{x}_r\ddot{y}_r - \dot{y}_r\ddot{x}_r}{v_r^3}, \qquad a_r = \frac{\dot{x}_r\ddot{x}_r + \dot{y}_r\ddot{y}_r}{v_r}, \tag{3.3}$$

$$\omega_r = v_r\kappa_r, \qquad \delta_r = \arctan(L\kappa_r). \tag{3.4}$$

Thus $\boldsymbol{u}_r = [\delta_r, a_r]^\top$ is the command that reproduces the reference in the nominal continuous model. Substitution gives $\dot{\psi}_r = v_r\tan\delta_r/L = \omega_r$ and $\dot{v}_r = a_r$. These identities ensure that zero tracking error with the feedforward command is a consistent nominal motion.

3.3 Numerical propagation

The controller updates at $T_s = 0.1$ s. The plant uses five RK4 substeps per interval, each of length $h = 0.02$ s. With a held command and derivative f from (3.1), one substep is

$$\boldsymbol{k}_1 = f(\boldsymbol{z}, t), \qquad \boldsymbol{k}_2 = f(\boldsymbol{z} + \tfrac{h}{2}\boldsymbol{k}_1, t + \tfrac{h}{2}), \tag{3.5}$$

$$\boldsymbol{k}_3 = f(\boldsymbol{z} + \tfrac{h}{2}\boldsymbol{k}_2, t + \tfrac{h}{2}), \qquad \boldsymbol{k}_4 = f(\boldsymbol{z} + h\boldsymbol{k}_3, (t+h)^-), \tag{3.6}$$

$$\boldsymbol{z}^+ = \boldsymbol{z} + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4). \tag{3.7}$$

The endpoint is evaluated from the left at scheduled disturbance boundaries. Commands are applied without clipping, allowing the recorded inputs to test the optimizer's constraints directly. Heading is wrapped after propagation.

4 Derivation of moving-frame tracking errors

Define a rotation from the global frame to the reference tangent-normal frame:

$$T(\psi_r) = \begin{bmatrix} \cos\psi_r & \sin\psi_r \\ -\sin\psi_r & \cos\psi_r \end{bmatrix}, \qquad \begin{bmatrix} e_s \\ e_y \end{bmatrix} = T(\psi_r)(\boldsymbol{p} - \boldsymbol{p}_r). \tag{4.1}$$

Positive e_s means the vehicle is ahead of the reference along its tangent; positive e_y means it lies to the reference's left. Define

$$e_\psi = \mathrm{wrap}(\psi - \psi_r), \qquad e_v = v - v_r. \tag{4.2}$$

The angle wrapping selects the local error in $[-\pi, \pi]$; the following differential equations apply away from the branch discontinuity.

4.1 Position-error dynamics

Writing $\boldsymbol{e}_p = [e_s, e_y]^\top$ and differentiating the rotating-frame expression yields

$$\dot{\boldsymbol{e}}_p = \dot{T}T^\top \boldsymbol{e}_p + T(\dot{\boldsymbol{p}} - \dot{\boldsymbol{p}}_r), \qquad \dot{T}T^\top = \begin{bmatrix} 0 & \omega_r \\ -\omega_r & 0 \end{bmatrix}. \tag{4.3}$$

The nominal vehicle velocity in this frame is $[v\cos e_\psi, v\sin e_\psi]^\top$, while reference velocity is $[v_r, 0]^\top$. Therefore the exact nominal error dynamics are

$$\dot{\boldsymbol{e}} = \begin{bmatrix} (v_r + e_v)\cos e_\psi - v_r + \omega_r e_y \\ (v_r + e_v)\sin e_\psi - \omega_r e_s \\ \dfrac{v_r + e_v}{L}\tan\delta - \omega_r \\ a - a_r \end{bmatrix}. \tag{4.4}$$

The terms $+\omega_r e_y$ and $-\omega_r e_s$ arise from frame rotation. Omitting them would change the model on curved trajectories. The position errors are not computed by searching for the nearest waypoint: the frame is anchored at the reference time.

4.2 How the disturbance enters these coordinates

The global disturbance transforms as $T[0, d_y]^\top$. The disturbed error model adds

$$\boldsymbol{w}_e(t) = \begin{bmatrix} \sin\psi_r\, d_y \\ \cos\psi_r\, d_y \\ 0 \\ d_a \end{bmatrix} \tag{4.5}$$

to (4.4). This expression explains why a global lateral disturbance need not act exclusively on cross-track error during a turn. The controller omits $\boldsymbol{w}_e$ and reacts to its effects through subsequent measured errors.

5 Reference-scheduled linear prediction model

5.1 Linearization about a reference motion

Linearize (4.4) at $\boldsymbol{e} = 0$ and $\boldsymbol{u} = \boldsymbol{u}_r$. With $\widetilde{\boldsymbol{u}} = \boldsymbol{u} - \boldsymbol{u}_r$, the local approximation is

$$\dot{\boldsymbol{e}} \simeq A_c(\rho)\boldsymbol{e} + B_c(\rho)\widetilde{\boldsymbol{u}}, \qquad \rho = [v_r, \kappa_r]^\top. \tag{5.1}$$

Since the nominal derivative vanishes at the expansion point, there is no independent affine term in these error coordinates. The Jacobians are

$$A_c = \begin{bmatrix} 0 & \omega_r & 0 & 1 \\ -\omega_r & 0 & v_r & 0 \\ 0 & 0 & 0 & \kappa_r \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{v_r}{L\cos^2\delta_r} & 0 \\ 0 & 1 \end{bmatrix}. \tag{5.2}$$

For example, the heading-error derivative with respect to speed error is $\tan\delta_r/L = \kappa_r$, and its derivative with respect to steering is $v_r \sec^2\delta_r/L$. These are distinct couplings: speed changes affect heading on a curved reference even when steering remains at its feedforward value.

5.2 Discretization and preview scheduling

Forward Euler yields

$$A_i = I + T_s A_c(\rho_{i|k}), \qquad B_i = T_s B_c(\rho_{i|k}), \tag{5.3}$$

$$\boldsymbol{e}_{i+1|k} = A_i \boldsymbol{e}_{i|k} + B_i(\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k}), \quad i = 0, \ldots, N-1. \tag{5.4}$$

The horizon is $NT_s = 2.0$ s. Each pair (A_i, B_i) is calculated from the corresponding future reference sample. The scheduling sequence is known before optimization; predicted states do not update it inside a nonlinear iteration.

5.3 Meaning of the LPV description

The model is linear in errors and command deviations, with coefficients depending on reference parameters. For a fixed preview it is a linear time-varying (LTV) prediction sequence. It is a *local reference-scheduled LPV approximation*, not an exact global LPV representation of the nonlinear vehicle. Euler discretization and neglected higher-order terms contribute prediction error. Separate RK4 plant integration ensures the evaluation does not merely propagate the same linear model used by the optimizer.

6 Physical limits and horizon constraints

The decision variables are absolute steering and acceleration commands. Their amplitude bounds are

$$\boldsymbol{u}_{\min} \le \boldsymbol{u}_{i|k} \le \boldsymbol{u}_{\max}, \quad \boldsymbol{u}_{\min} = \begin{bmatrix} -25^\circ \\ -2.5 \end{bmatrix}, \quad \boldsymbol{u}_{\max} = \begin{bmatrix} 25^\circ \\ 2.0 \end{bmatrix}. \tag{6.1}$$

All angular quantities are converted to radians in the implementation. Acceleration is in $\mathrm{m/s^2}$.

6.1 Rate limits and the previously applied command

Define $\Delta\boldsymbol{u}_{0|k} = \boldsymbol{u}_{0|k} - \boldsymbol{u}_{k-1}$ and $\Delta\boldsymbol{u}_{i|k} = \boldsymbol{u}_{i|k} - \boldsymbol{u}_{i-1|k}$ for $i \ge 1$. The inequalities are

$$-\boldsymbol{r}T_s \le \Delta\boldsymbol{u}_{i|k} \le \boldsymbol{r}T_s, \qquad \boldsymbol{r} = \begin{bmatrix} 35^\circ/\mathrm{s} \\ 3.0\ \mathrm{m/s^3} \end{bmatrix}. \tag{6.2}$$

The acceleration-rate constraint bounds *commanded jerk*. It does not bound the derivative of total physical acceleration when an external load changes abruptly. At the first sample $\boldsymbol{u}_{-1} = 0$, limiting initial changes to 3.5° steering and $0.3\ \mathrm{m/s^2}$ acceleration per interval. This prevents the controller from instantly applying a large correction to the initial error.

6.2 Predicted-speed bounds

Because $v_{i|k} = v_{r,i|k} + e_{v,i|k}$, the limits are linear in the error state:

$$1 - v_{r,i|k} \le e_{v,i|k} \le 10 - v_{r,i|k}, \qquad i = 1, \ldots, N. \tag{6.3}$$

These apply to predicted future samples. The initial measured error is imposed as an equality; speed at sample zero is not an additional QP bound. The saved nonlinear plant trajectory is checked separately for sampled speed violations.

6.3 Why feasibility matters

All constraints are hard; there are no slack variables. A previously applied command far outside the amplitude range can make it impossible to re-enter that range within one allowed rate step. For instance, a prior steering of 2 rad cannot reach $25^\circ \simeq 0.4363$ rad with a maximum step of approximately 0.0611 rad. The numerical test suite deliberately constructs this infeasible case.

Constraint satisfaction for the predicted model does not automatically establish constraint satisfaction between samples or under arbitrary plant mismatch. The current recorded runs include independent sampled checks; no collision or road-boundary constraints are present.

7 Objective function and tuning interpretation

At sample k, the finite-horizon objective is

$$\begin{aligned} J_k = & \sum_{i=0}^{N-1} \boldsymbol{e}_{i|k}^\top Q \boldsymbol{e}_{i|k} + \boldsymbol{e}_{N|k}^\top P_f \boldsymbol{e}_{N|k} \\ & + \sum_{i=0}^{N-1} (\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k})^\top R (\boldsymbol{u}_{i|k} - \boldsymbol{u}_{r,i|k}) \\ & + \sum_{i=0}^{N-1} \Delta \boldsymbol{u}_{i|k}^\top S \Delta \boldsymbol{u}_{i|k}. \end{aligned} \tag{7.1}$$

The selected matrices are

$$Q = \mathrm{diag}(4, 22, 12, 4), \quad R = \mathrm{diag}(3, 0.25), \quad S = \mathrm{diag}(12, 0.8), \quad P_f = 3Q. \tag{7.2}$$

The implemented state weights act on meters, radians, and m/s. Their numerical sizes should not be compared as if all errors had the same units or scale.

7.1 What each term does

- **Tracking error.** The Q term rewards agreement with the reference motion. Cross-track error is emphasized, while along-track and speed error remain explicit objectives.
- **Feedforward deviation.** The R term penalizes departure from the nominal steering and acceleration required by the reference. On a curve, correct steering is generally nonzero; penalizing deviation from δ_r avoids treating all steering as unnecessary effort.
- **Command smoothness.** The S term discourages rapid command changes even when they remain inside the hard rate limits. Rate bounds define admissibility; the rate penalty ranks admissible sequences.
- **Terminal tracking.** $P_f = 3Q$ adds weight to the final error. It is a design choice, not a Riccati-derived terminal certificate or an invariant terminal-set construction.

7.2 Interpretation of trade-offs

Increasing tracking weights tends to favor faster correction, potentially driving commands toward their bounds. Increasing rate penalties tends to smooth commands, potentially extending recovery. The actual trade-off depends on state units, reference curvature, sampling, constraints, and horizon length. No tuning sweep was performed for this report; these are interpretations of the objective, not measured comparisons between alternative settings.

The current measured error $\boldsymbol{e}_{0|k}$ is fixed, so its quadratic term contributes a constant to the optimization. Retaining it in the stacked objective simplifies the formulation without changing the minimizing command sequence.

8 Sparse quadratic-program construction

Define the stacked error and command vectors

$$\boldsymbol{E} = \begin{bmatrix} \boldsymbol{e}_{0|k} \\ \vdots \\ \boldsymbol{e}_{N|k} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} \boldsymbol{u}_{0|k} \\ \vdots \\ \boldsymbol{u}_{N-1|k} \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{E} \\ \boldsymbol{U} \end{bmatrix} \in \mathbb{R}^{124}. \tag{8.1}$$

There are $4(N+1) = 84$ error-state entries and $2N = 40$ command entries. Define

$$\bar{Q} = \operatorname{blkdiag}(Q, \ldots, Q, P_f), \quad \bar{R} = I_N \otimes R, \quad \bar{S} = I_N \otimes S, \tag{8.2}$$

$$D = \begin{bmatrix} I_2 & 0 & \cdots & 0 \\ -I_2 & I_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & -I_2 & I_2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} \boldsymbol{u}_{k-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{8.3}$$

Thus $D\boldsymbol{U} - \boldsymbol{b}$ is the stacked command-increment vector. Expanding (7.1) and dropping constants gives the solver form

$$\min_{\boldsymbol{\xi}} \ \tfrac{1}{2}\boldsymbol{\xi}^\top H \boldsymbol{\xi} + \boldsymbol{q}^\top \boldsymbol{\xi} \quad \text{subject to} \quad \boldsymbol{\ell} \le C\boldsymbol{\xi} \le \boldsymbol{h}, \tag{8.4}$$

where

$$H = 2\operatorname{blkdiag}(\bar{Q}, \bar{R} + D^\top \bar{S} D), \quad \boldsymbol{q} = \begin{bmatrix} 0 \\ -2(\bar{R}\boldsymbol{U}_r + D^\top \bar{S}\boldsymbol{b}) \end{bmatrix}. \tag{8.5}$$

The factor of two follows from the solver's $\frac{1}{2}\boldsymbol{\xi}^\top H \boldsymbol{\xi}$ convention. Omitting it from only one objective component would change the relative weights.

8.1 Dynamics and inequality blocks

The first four constraint rows impose $\boldsymbol{e}_{0|k} = \boldsymbol{e}_k$. For each step, the equality row is

$$-A_i \boldsymbol{e}_{i|k} + \boldsymbol{e}_{i+1|k} - B_i \boldsymbol{u}_{i|k} = -B_i \boldsymbol{u}_{r,i|k}. \tag{8.6}$$

Equality rows use identical lower and upper bounds. The remaining blocks select $\boldsymbol{U}$, $D\boldsymbol{U}$, and future speed errors. In total, $C \in \mathbb{R}^{184\times 124}$ has 84 equality rows, 40 amplitude rows, 40 rate rows, and 20 speed rows. Keeping states explicit produces a sparse dynamics structure rather than eliminating them into a dense command-only problem.

9 Optimization, feedback, and numerical acceptance

9.1 Receding-horizon procedure

1. At t_k, calculate $\boldsymbol{e}_k$ from the measured vehicle state and reference.
2. Obtain $N+1$ reference samples; construct N scheduled model pairs and nominal commands.
3. Assemble $\boldsymbol{q}$, C, $\boldsymbol{\ell}$, and $\boldsymbol{h}$, incorporating the previously applied command.
4. Solve (8.4), check solver status, finite values, and constraint residuals.
5. Apply only $\boldsymbol{u}^{\star}_{0|k}$ for one interval; integrate the nonlinear plant with RK4.
6. Measure the new state and repeat with a shifted reference window.

Feedback comes from repeatedly replacing the initial predicted error by the measured error. A disturbance creates a discrepancy between the previous prediction and the next measurement, which triggers a new corrective solution.

9.2 Numerical solver

OSQP solves the convex QP using an operator-splitting method based on the alternating direction method of multipliers (ADMM) [2, 3]. The implementation requests absolute and relative tolerances of 10^{-6}, permits 12,000 iterations, and enables polishing. Only the `solved` status is accepted. A fresh solver is created at every control step; the previous primal solution is supplied as a warm start. The implementation does not reuse a persistent factorization across steps or explicitly shift the previous trajectory before warm starting.

An independent feasibility check computes

$$\epsilon_c = \max\left(0, \max_j[\ell_j - (C\boldsymbol{\xi})_j], \max_j[(C\boldsymbol{\xi})_j - h_j]\right). \tag{9.1}$$

A solution is rejected if any entry is nonfinite or $\epsilon_c > 2 \times 10^{-5}$. This mixed-unit maximum is an implementation acceptance threshold, not a normalized physical safety margin. Failure raises an error; a fallback policy is not implemented.

9.3 Convexity and solution uniqueness

Every diagonal entry of Q, P_f, and R is positive, while S is positive definite. Hence $\bar{R} + D^{\top}\bar{S}D \succ 0$ and $H \succ 0$. All constraints are affine for a fixed reference preview. Consequently, a feasible finite QP has a unique minimizing decision vector. This is a property of the prediction problem; it does not establish global optimality of the nonlinear closed-loop motion.

10 Local control properties and limits of the theory

10.1 Straight-reference controllability

For $\kappa_r = 0$ and constant $v_r > 0$, the continuous local model separates into two second-order chains. The longitudinal pair (e_s, e_v) satisfies

$$\frac{d}{dt}\begin{bmatrix} e_s \\ e_v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} e_s \\ e_v \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\widetilde{a}. \tag{10.1}$$

Its controllability matrix $[B, AB]$ has determinant -1. The lateral pair (e_y, e_ψ) satisfies

$$\frac{d}{dt}\begin{bmatrix} e_y \\ e_\psi \end{bmatrix} = \begin{bmatrix} 0 & v_r \\ 0 & 0 \end{bmatrix}\begin{bmatrix} e_y \\ e_\psi \end{bmatrix} + \begin{bmatrix} 0 \\ v_r/L \end{bmatrix}\widetilde{\delta}. \tag{10.2}$$

Its controllability determinant is $-v_r^3/L^2$, nonzero for $v_r > 0$. This local calculation explains the steering authority in the model and its loss at zero speed. It is not a controllability proof for every scheduled, constrained, disturbed trajectory.

10.2 Model mismatch and disturbance rejection

The nonlinear error system can be written locally as

$$\dot{\boldsymbol{e}} = A_c\boldsymbol{e} + B_c\widetilde{\boldsymbol{u}} + \boldsymbol{r}_{\mathrm{nl}}(\boldsymbol{e}, \widetilde{\boldsymbol{u}}, \rho) + \boldsymbol{w}_e, \tag{10.3}$$

where $\boldsymbol{r}_{\mathrm{nl}}$ collects terms neglected by linearization. Prediction additionally incurs discretization error. The controller corrects their measured effects through feedback, but neither a disturbance observer nor integral-error states are present. Zero steady-state error under an arbitrary persistent disturbance therefore does not follow from this formulation.

10.3 What is not established by the terminal weight

A terminal penalty $P_f = 3Q$ encourages small horizon-end error. It does not supply the terminal invariant set, terminal policy, or decrease condition typically used for a recursive-feasibility or stability proof [1]. There is no tube tightening, min–max formulation, or robust invariant-set argument in this implementation. The report therefore distinguishes convexity of each QP, numerical feasibility of recorded runs, and closed-loop stability: these are different properties.

The available evidence supports recovery for the prescribed finite experiments. Extending that conclusion to larger disturbances, uncertain vehicle parameters, measurement errors, or high-slip maneuvers requires additional modeling and tests.

11 Experiment design and performance measures

11.1 Analytic reference trajectories

The three reference curves are defined directly in time:

$$\text{Oval:} \quad x_r = 42\cos\theta, \quad y_r = 25\sin\theta, \quad \theta = 2\pi t/52, \tag{11.1}$$

$$\text{Figure-eight:} \quad x_r = 45\sin\theta, \quad y_r = 16\sin 2\theta, \quad \theta = 2\pi t/64, \tag{11.2}$$

$$\text{Lane change:} \quad x_r = 5t, \quad y_r = 3\left[\tanh\frac{t-9}{2} - \tanh\frac{t-19}{2}\right]. \tag{11.3}$$

The lane-change expression produces a lateral excursion of approximately 6 m and a return, rather than assuming a standard lane width. The oval changes both reference speed and curvature; the figure-eight reverses curvature and crosses itself. Time-indexed tracking removes ambiguity at that crossing.

Each run begins with the same reference-frame error and previous command:

$$\boldsymbol{e}_0 = [-1\text{ m},\ 1.4\text{ m},\ 8^\circ,\ -0.7\text{ m/s}]^\top, \qquad \boldsymbol{u}_{-1} = 0. \tag{11.4}$$

Only the lane-change plant receives $d_y = 0.35$ m/s and $d_a = -0.6$ m/s^2 during $24 \le t < 26$ s. These values are not included in the controller's preview. Durations are 52, 64, and 36 s, producing 520, 640, and 360 optimizations respectively. Each trajectory includes the final state sample, for 1,523 state samples in total.

11.2 Metrics and sampling conventions

For a run containing M state samples,

$$\text{RMSE}_p = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1}[(x_j - x_{r,j})^2 + (y_j - y_{r,j})^2]}, \tag{11.5}$$

$$\text{RMSE}_y = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1} e_{y,j}^2}, \qquad \text{RMSE}_v = \sqrt{\frac{1}{M}\sum_{j=0}^{M-1} e_{v,j}^2}. \tag{11.6}$$

Full-run values include startup recovery and the disturbance. Orthogonality of the frame transformation gives $(x - x_r)^2 + (y - y_r)^2 = e_s^2 + e_y^2$. Position RMSE therefore includes along-track timing error; it is not interchangeable with cross-track RMSE.

The stored record also contains cross-track RMSE restricted to $t \ge 5$ s. Those restricted values are discussed only as supplemental diagnostics and are not substituted for the primary full-run results. Control-rate checks include the transition from the zero initial command.

12 Trajectory tracking results

Scenario	Position RMSE [m]	Cross-track RMSE [m]	Speed RMSE [m/s]
Variable-speed oval	0.3180	0.2498	0.1206
Figure-eight	0.2893	0.1966	0.1246
Disturbed lane change	0.3958	0.2708	0.1707

Table 1: Recomputed full-run errors from the saved state and reference arrays.

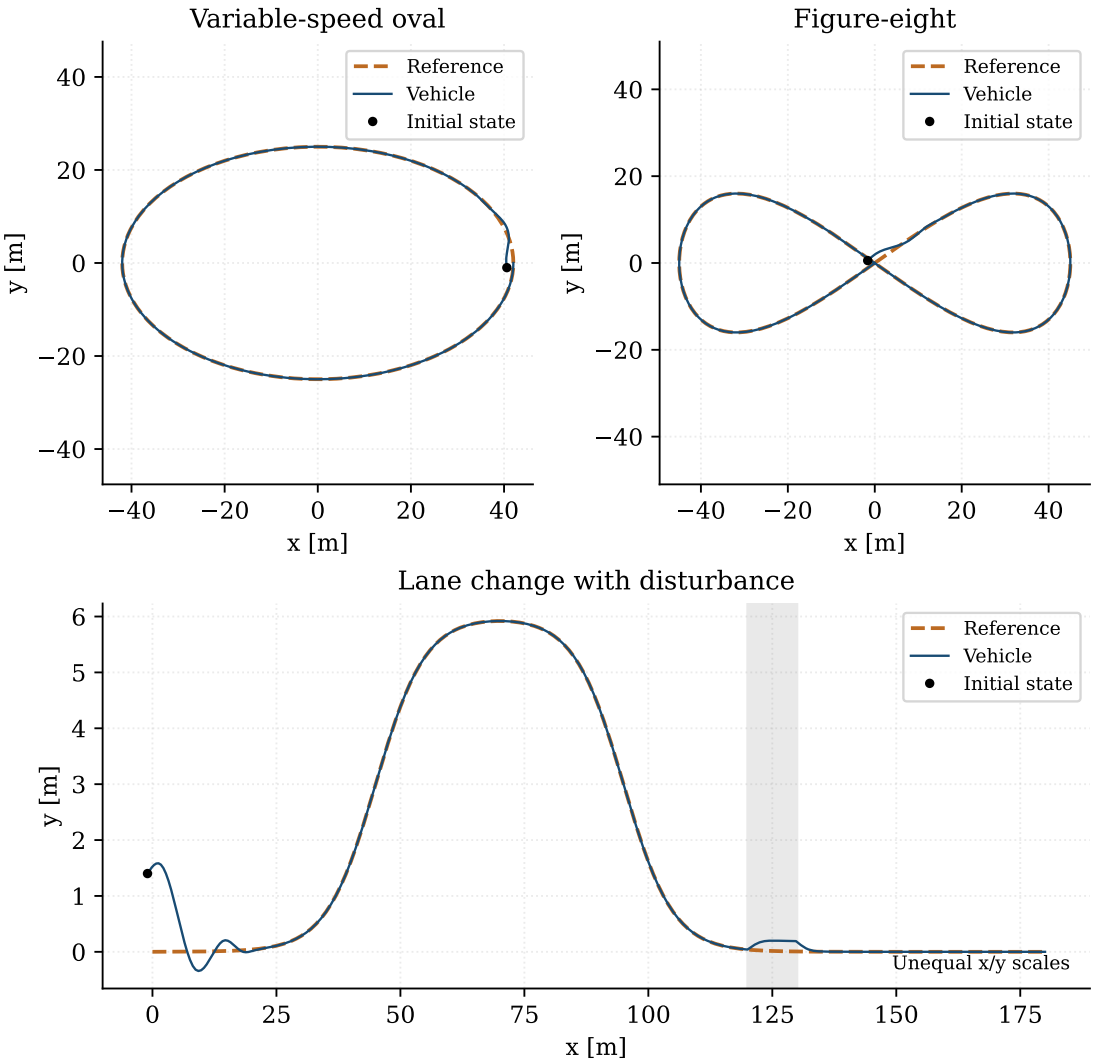


Figure 2: Reference and simulated trajectories. Initial position offsets are included. The shaded lane-change region corresponds to reference travel during the disturbance interval; that panel uses unequal horizontal and vertical scales.

The three runs show convergence toward the reference after the initial offset. Maximum position errors are 1.935, 2.038, and 2.034 m respectively. Those peaks exceed the initial position error $\sqrt{1^2 + 1.4^2} \simeq 1.720$ m, so the transient should not be described as monotonic error reduction. The initial heading error and command-rate limits matter during recovery.

13 Error transients and interpretation

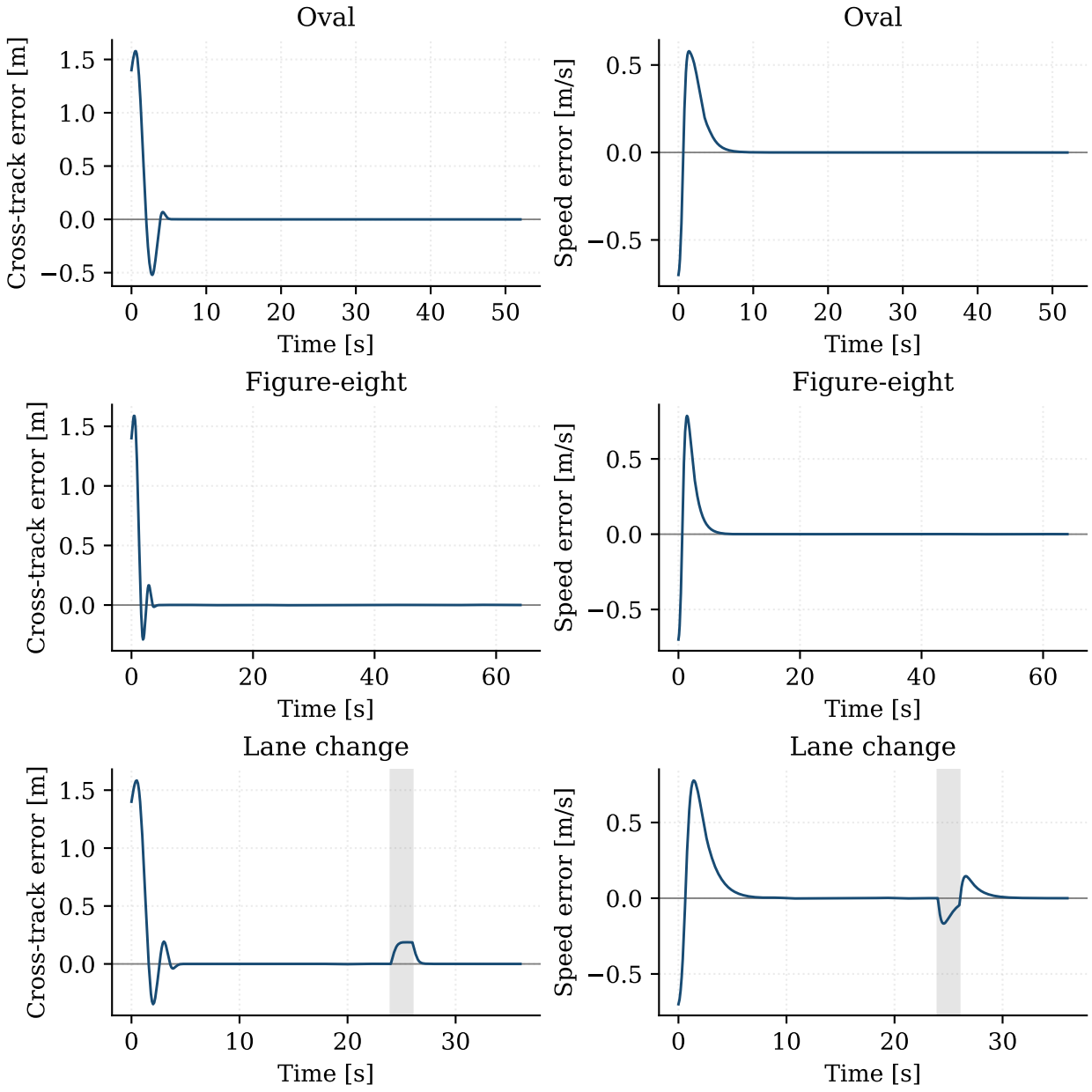


Figure 3: Signed cross-track and speed errors over all three runs. Gray shading marks the lane-change disturbance at 24–26 s. Startup remains visible in every panel.

After the first five seconds, recorded cross-track RMSE is 0.000587 m on the oval and 0.000935 m on the figure-eight. These small values arise under ideal state feedback, an exact analytic reference, and the specified kinematic plant. They should not be interpreted as hardware localization or tracking accuracy.

For the lane change, the corresponding $t \geq 5$ s value is 0.0441 m, including the later disturbance. This separates the effect of startup from the remainder of the run without removing the disturbance. The higher full-run lane-change errors are consistent with the imposed load, drift, and different reference geometry, but the study has no matched undisturbed lane-change baseline from which to isolate a causal performance penalty.

14 Disturbance response and recovery analysis

The temporary disturbance is injected into the nonlinear plant only. Positive global lateral drift initially moves the vehicle away from its reference, while negative longitudinal load reduces speed. The next optimization receives the resulting tracking error and adjusts steering and acceleration.

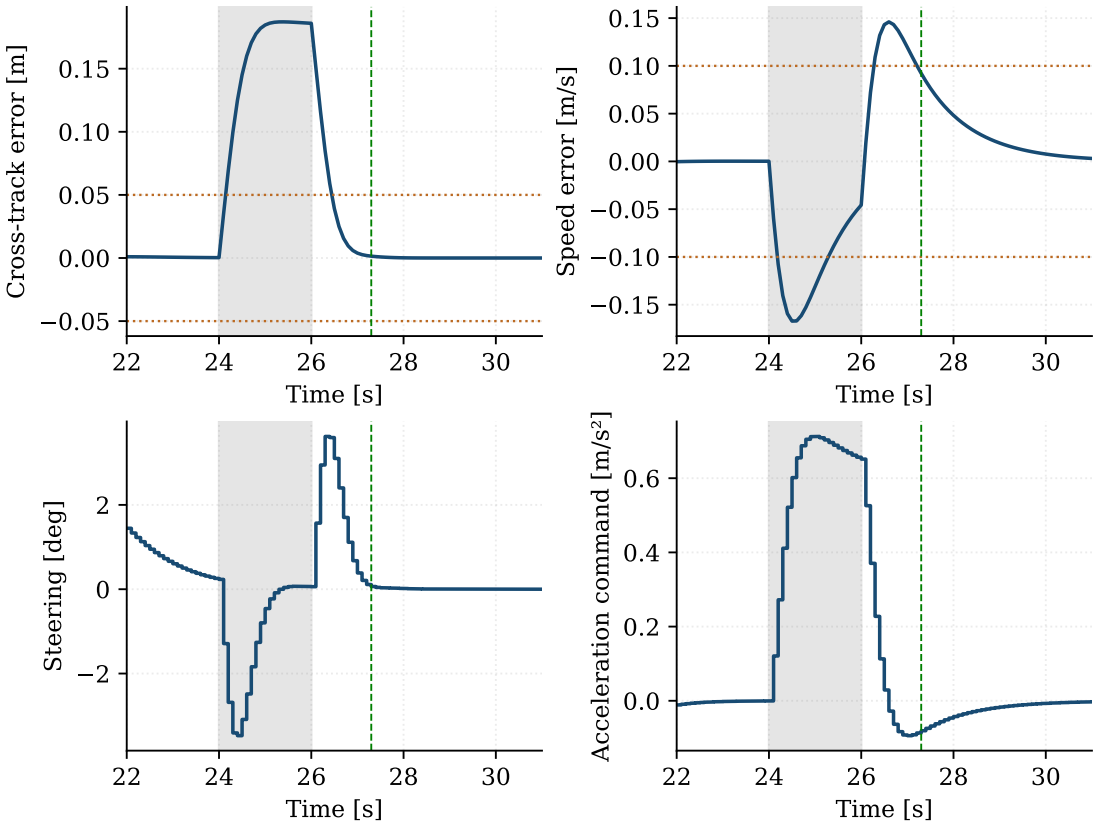


Figure 4: Saved lane-change response around the disturbance. Gray shading denotes 24–26 s. Dotted orange error thresholds define the report's recovery criterion; the green line denotes its first sustained sampled satisfaction at 27.3 s.

Quantity	Value
Maximum $\lvert e_y \rvert$ during $24 \le t < 26$ s	0.1871 m
Maximum $\lvert e_v \rvert$ during $24 \le t < 26$ s	0.1672 m/s
Cross-track error at 26 s	0.1860 m
Speed error at 26 s	−0.0457 m/s
First sustained threshold satisfaction after removal	27.3 s
Delay from disturbance removal to that sample	1.3 s

For this report, recovery is defined as the first sampled time at or after 26 s for which $\lvert e_y \rvert \le 0.05$ m and $\lvert e_v \rvert \le 0.10$ m/s remain satisfied through the end of the saved run. This is a post-processing criterion chosen for analysis, not a controller requirement. It does not test along-track or heading settling, and the 0.1 s sampling limits its time resolution. The result demonstrates finite-run recovery from this specific temporary disturbance, not a bound for all disturbances.

15 Command limits and optimization outcomes

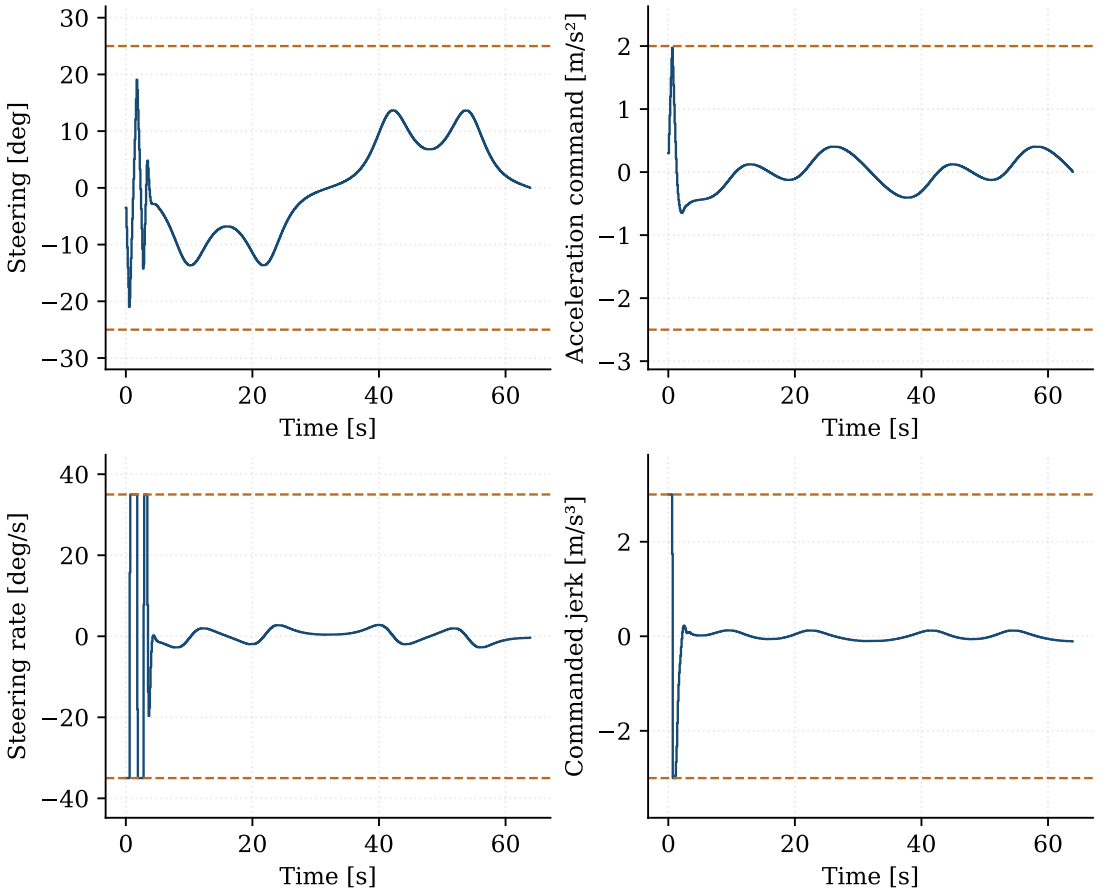


Figure 5: Figure-eight commands and their sample-to-sample rates. Dashed lines show configured limits. Rate saturation reflects active constraints rather than post-processing clipping.

Recorded quantity	Oval	Figure-eight	Lane change
Peak $\lvert\delta\rvert$ [deg]	25.000	21.000	21.265
Peak $\lvert\Delta\delta\rvert/T_s$ [deg/s]	35.000	35.000	35.000
Peak $\lvert\Delta a\rvert/T_s$ [m/s^3]	3.000	3.000	3.000
Acceleration range [m/s^2]	$[-0.248, 1.869]$	$[-0.647, 1.974]$	$[-0.417, 2.000]$
Speed range [m/s]	$[2.321, 5.075]$	$[2.710, 6.107]$	$[4.300, 5.778]$
Successful QP solves	520	640	360

All saved command, rate, and sampled-speed checks pass their configured tolerances. The largest recorded QP constraint residual is approximately 7.16×10^{-12} across the three runs. Tiny floating-point excursions around nominal limits are evaluated using the implementation's tolerances.

Recorded median controller times are approximately 0.899, 0.938, and 0.937 ms, with 95th-percentile times of 1.279, 1.075, and 1.403 ms. These stored local timings include QP assembly, setup, and solve, exclude plant integration, and lack a documented hardware specification. They are diagnostic measurements, not a transferable real-time benchmark or deadline guarantee.

16 Verification and reproducibility

16.1 Evidence inspected for this report

The mathematical description was checked against the model, QP assembly, simulation loop, and test definitions. SHA-256 hashes for all five Python source files match the hashes stored with the original experiment metrics. This links the inspected implementation to the recorded results without rerunning the complete experiments.

The report preparation independently recomputed full-run position, cross-track, and speed RMSE from the saved arrays. It reconstructed tracking errors from state/reference pairs, checked sampled input, rate, and speed bounds, and calculated the disturbance-window statistics and recovery time. The analytical Jacobians were compared against central finite differences at three figure-eight reference points, agreeing within 10^{-7} absolute tolerance.

16.2 Numerical checks present in the project

Check	Property examined
Reference derivatives	Analytic reference velocity and acceleration agree with numerical differentiation.
Local error Jacobians	Scheduled A_c and B_c agree with finite differences of the nonlinear error model.
Constant-turn integration	RK4 follows an analytically known circle under constant steering and speed.
Coordinate conventions	Reference-frame rotation and angle wrapping produce expected signed errors.
Equilibrium / infeasibility	Straight nominal motion yields zero commands; an incompatible prior steering command is rejected.

These are the five checks implemented in the original test module. The report preparation reran the Jacobian check and the saved-data checks described above; it did not rerun the full solver test suite or launch a new simulation. This distinction keeps historical experiment evidence separate from the checks performed while preparing the document.

16.3 Stored numerical environment

The saved experiment record identifies NumPy 2.2.6, SciPy 1.15.3, OSQP 1.1.3, Matplotlib 3.10.9, and Pillow 12.3.0. The solver settings and physical limits are fixed in the recorded configuration. Raw state, reference, error, command, and timing arrays accompany each scenario. No random disturbances or measurement-noise seeds are involved.

The report's figures are regenerated from those saved arrays, not traced from screenshots. The controller source remains outside the public website; the report presents equations and measured outcomes without distributing implementation code.

17 Comparison, contribution, and research extensions

Aspect	Supplied reference study	Present study
Broad objective	Vehicle trajectory tracking	Constrained tracking and recovery evaluation
Vehicle model	Six states; lateral velocity and yaw-rate dynamics	Four-state kinematic bicycle
Tracking representation	Speed, heading, and global position outputs	Moving-frame position, heading, and speed errors
Prediction horizon	10 steps at 0.02 s; 0.2 s	20 steps at 0.1 s; 2.0 s
Optimization	Condensed formulation with CVXOPT	Explicit state/command sparse QP with OSQP
Recorded evaluation	Supplied default trajectory replay	Oval, figure-eight, disturbed lane change

The current study's defensible contribution is its formulation and experiment design: it tests how a reference-scheduled controller handles changing geometry, initial misalignment, and a temporary disturbance under explicit command limits. The controller is implemented separately from the supplied reference, while retaining the same broad MPC principle. A simpler kinematic model is a modeling choice, not evidence of greater physical fidelity. Different horizons, trajectories, and models prevent a direct numerical superiority claim.

17.1 Extensions that would answer stronger questions

1. **Matched controller comparison.** Evaluate fixed-linear and scheduled-linear MPC on the same plant, trajectories, constraints, initial conditions, and disturbance. This would isolate the effect of scheduling.
2. **Tuning and constraint ablation.** Vary horizon length and rate penalties, measuring recovery time, full-run error, constraint activation, and computational effort. This would quantify the tracking–smoothness trade-off.
3. **Persistent-disturbance estimation.** Add an offset-free formulation or an augmented disturbance model, then test sustained loads. This would address residual offsets beyond temporary-disturbance recovery.
4. **Model uncertainty and estimation.** Introduce tire dynamics, parameter changes, delayed/noisy state estimates, and actuator lag, using matched baselines. This would assess applicability beyond the ideal kinematic setting.

These are proposed extensions, not completed results. The present work establishes a traceable mathematical and numerical basis for them.

A Worked local model and QP dimensions

Consider a reference point with $v_r = 4$ m/s, $\kappa_r = 0.02$ m^{-1}, and $L = 2.7$ m. Then

$$\omega_r = 0.08 \text{ rad/s}, \qquad \delta_r = \arctan(0.054) \simeq 0.05395 \text{ rad}. \tag{A.1}$$

Since $\sec^2(\arctan q) = 1 + q^2$, the steering gain is

$$g_\delta = \frac{4}{2.7}(1 + 0.054^2) \simeq 1.48580 \text{ s}^{-1}. \tag{A.2}$$

At $T_s = 0.1$ s, the prediction matrices become

$$A_d = \begin{bmatrix} 1 & 0.008 & 0 & 0.1 \\ -0.008 & 1 & 0.4 & 0 \\ 0 & 0 & 1 & 0.002 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.14858 & 0 \\ 0 & 0.1 \end{bmatrix}. \tag{A.3}$$

For $\boldsymbol{e} = [-1, 1.4, 0.139626, -0.7]^\top$ and zero command deviation from feedforward,

$$\boldsymbol{e}^+ \simeq A_d \boldsymbol{e} = [-1.0588,\ 1.46385,\ 0.138226,\ -0.7]^\top. \tag{A.4}$$

Thus nominal feedforward alone does not remove the offset. The positive cross-track increment is consistent with the heading error and frame-rotation terms. A feedback optimization must choose command deviations to alter this evolution.

A.1 Horizon structure

For $N = 20$, the decision vector contains 21 four-dimensional error states and 20 two-dimensional commands. A single dynamics equality couples only $\boldsymbol{e}_i$, $\boldsymbol{e}_{i+1}$, and $\boldsymbol{u}_i$. The difference operator couples adjacent commands. This block-local structure explains the sparse assembly.

With only two predicted intervals, the dynamics rows would have the form

$$\begin{bmatrix} I & 0 & 0 & 0 & 0 \\ -A_0 & I & 0 & -B_0 & 0 \\ 0 & -A_1 & I & 0 & -B_1 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_0 \\ \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \boldsymbol{u}_0 \\ \boldsymbol{u}_1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{e}_k \\ -B_0 \boldsymbol{u}_{r,0} \\ -B_1 \boldsymbol{u}_{r,1} \end{bmatrix}. \tag{A.5}$$

The full horizon repeats this pattern. Absolute commands appear in the decision vector, while reference commands appear on the right-hand side and in the objective's linear term.

B Optimality conditions and theoretical interpretation

Rewrite (8.4) with equality constraints $A_{\text{eq}}\boldsymbol{\xi} = \boldsymbol{b}_{\text{eq}}$ and inequalities $G\boldsymbol{\xi} \le \boldsymbol{g}$. For equality multipliers $\boldsymbol{\lambda}$ and nonnegative inequality multipliers $\boldsymbol{\mu}$, the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\boldsymbol{\xi}^\top H\boldsymbol{\xi} + \boldsymbol{q}^\top\boldsymbol{\xi} + \boldsymbol{\lambda}^\top(A_{\text{eq}}\boldsymbol{\xi} - \boldsymbol{b}_{\text{eq}}) + \boldsymbol{\mu}^\top(G\boldsymbol{\xi} - \boldsymbol{g}). \tag{B.1}$$

The Karush–Kuhn–Tucker conditions are

$$H\boldsymbol{\xi}^\star + \boldsymbol{q} + A_{\text{eq}}^\top\boldsymbol{\lambda}^\star + G^\top\boldsymbol{\mu}^\star = 0, \tag{B.2}$$

$$A_{\text{eq}}\boldsymbol{\xi}^\star = \boldsymbol{b}_{\text{eq}}, \tag{B.3}$$

$$G\boldsymbol{\xi}^\star \le \boldsymbol{g}, \qquad \boldsymbol{\mu}^\star \ge 0, \tag{B.4}$$

$$\mu_j^\star(G\boldsymbol{\xi}^\star - \boldsymbol{g})_j = 0 \quad \text{for every } j. \tag{B.5}$$

These conditions express a balance between the objective gradient and the active constraints. When a rate limit is active, the optimizer cannot improve tracking simply by increasing that command change. It must redistribute the correction across later steps and possibly other commands. This provides a theoretical interpretation of the saturated rate traces.

For this strictly convex QP with affine constraints, a feasible optimum is unique, and multipliers satisfying these conditions certify its optimality. Numerical convergence and feasibility are evaluated to tolerance. The controller does not store dual histories in the published result arrays, so the report does not infer multiplier values or quantify constraint shadow prices.

B.1 The distinction between optimization and stability

The optimal value $V_N(\boldsymbol{e}_k)$ is a finite-horizon cost for the current reference preview and prior command. To use it as a Lyapunov function, one would need additional conditions establishing a decrease along the actual closed loop, accounting for moving references, hard constraints, model error, and disturbances. Strict convexity alone does not provide that decrease.

Similarly, feasibility at sample k is not by itself a recursive-feasibility proof. After a disturbed transition, the next measured initial condition can differ from the previously predicted one. A theoretical guarantee would require an appropriate invariant construction, uncertainty treatment, or another explicit argument. The recorded 1,520 solved QPs show numerical feasibility on the evaluated trajectories, with no broader guarantee inferred.

C Implemented components and correspondence to the formulation

The implementation combines a reference generator, a scheduled prediction model, a constrained optimizer, and a separate nonlinear simulation loop. This appendix identifies the completed components and connects them to the equations and results presented in the report.

Implemented component	What the implementation does
Analytic reference generation	Generates position, heading, speed, curvature, feedforward acceleration, and steering for the oval, figure-eight, and lane-change maneuvers. Supplies a continuous preview beyond each run's endpoint.
Tracking coordinates	Rotates global position differences into along-track and cross-track errors; computes wrapped heading error and speed error.
Scheduled prediction	Forms the analytical local error Jacobians from reference speed and curvature. Uses forward Euler at 0.1 s and updates all 20 prediction steps from the preview.
Sparse QP assembly	Stacks 84 error-state variables and 40 command variables. Constructs the tracking, feedforward, and command-rate cost, plus 184 rows of dynamics and bound constraints.
Optimization and command application	Solves with OSQP, warm-starts from the previous primal solution, checks status and feasibility, and applies only the first command pair.
Nonlinear simulation	Integrates the four-state kinematic bicycle with five RK4 substeps per control interval. Adds global lateral drift and longitudinal load to the lane-change plant during 24–26 s.
Recorded evaluation	Saves states, references, tracking errors, applied commands, solver timing, and summary metrics. Checks sampled command, rate, and speed limits and stores source hashes with the results.

Implemented behavior and reported analysis

The controller uses a known reference and measured state at every step; it does not implement a state estimator, a disturbance observer, or an obstacle planner. The reported 1,520 successful QP solves and full-run errors come from the saved execution records of the implemented simulation loop.

The 1.3 s disturbance-recovery result is computed afterward from those saved arrays using the explicit cross-track and speed thresholds in Section 14. It is an analysis result, not a termination condition or a recovery-time constraint inside the controller. The theoretical derivations explain the implemented model and objective; the discussion of future extensions remains separate from completed work.

References

[1] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., Nob Hill Publishing. Author-maintained book page: https://sites.engineering.ucsb.edu/~jbraw/mpc/. Background for finite-horizon control and the distinction between terminal design and stability guarantees.

[2] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An Operator Splitting Solver for Quadratic Programs," *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020. Author publication page: https://web.stanford.edu/~boyd/papers/osqp.html.

[3] OSQP developers, "The solver," official documentation. https://osqp.org/docs/solver/. Solver problem form, operator splitting, convergence checks, and polishing.

Evidence provenance

The experiment record contains five source hashes and three saved trajectory datasets. The report preparation matched those hashes, recomputed the headline errors, checked stored command/rate/speed limits, and generated the figures from the saved arrays. The disturbance recovery statistic is additional analysis of the existing lane-change data. No complete experiment was rerun for this report.

The analytical sections derive the equations implemented in the local project. External references identify the established MPC framework and numerical solver; they are not evidence for this project's measured performance. No source listing or downloadable controller archive is included in the report.

Project statement

Investigated constrained vehicle tracking and disturbance recovery using a reference-scheduled LPV-MPC controller, evaluating tracking accuracy and command smoothness across three driving scenarios. The study combines a moving-frame prediction model, sparse constrained optimization, and nonlinear plant evaluation to examine recovery from initial offsets and a temporary unmodeled disturbance.